

Pocketful — Autonomous Zero-Sum Fintech Engine

Decoupled 3-Seat Autonomous Factory & Bitemporal Ledger Architecture

Mathematical Invariant

Strict zero-sum conservation across accounts, escrow, and balance reservations: $\text{Sum}(\text{Delta Balances}) = 0$.

Cryptographic Audit Chain

Tamper-proof ledger logging using chained SHA-256 hashes with UTC ISO-8601 evidentiary timestamps.

AML Sentinel Engine

Real-time velocity anomaly detection, micro-structuring heuristics, and multi-party fraud deterrence.

AUTONOMOUS EXECUTION (50% RUBRIC)

Decoupled 3-Seat Multi-Agent Topology

Zero human-in-the-loop steering inside BAND Desktop

1. Strategic Planner

Translates generic mandates into granular architectural specifications, interface contracts, and state boundaries.

2. Senior Implementer

Executes clean-room Python/FastAPI bitemporal transaction models with mutex locks for concurrency control.

3. Lead QA Reviewer

Adversarial fuzzing, zero-sum invariant stress tests, and automated continuous regression verification.

Formal Testing Matrix & Production Health

193 / 193 Contract Tests Passed | 0 Harness Problems

193 / 193

Test Contracts Passed

Continuous progression across Stages 1, 2, 3, and 4 without regression.

0 Problems

Harness Gate Status

Passed all integrity gates: Gate 1 (Structure), Gate 2 (Contract), Gate 4 (Deployment).

100%

Zero-Network Container

Multi-stage Dockerfile runnable in isolated air-gapped environments and deployed to Vercel Edge.